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NI-NLM – Lecture 2

Tokenization & the Transformer architecture

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Recap & math refresher

Recap from last lecture

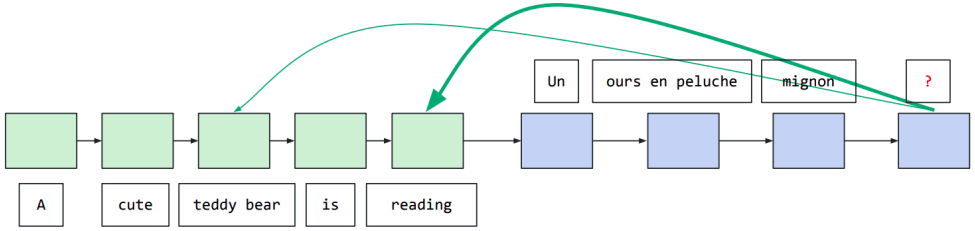
Last time, we covered:

- How to represent **language**?
- What is language **modeling**?
- How to process language with **neural networks**?
  - Also how to do language modeling with neural networks.

Today, we will cover:

How to process language in a more **efficient and scalable way**?

(→ tokenization, attention, Transformer)



source: Stanford CME295

Matrix multiplication

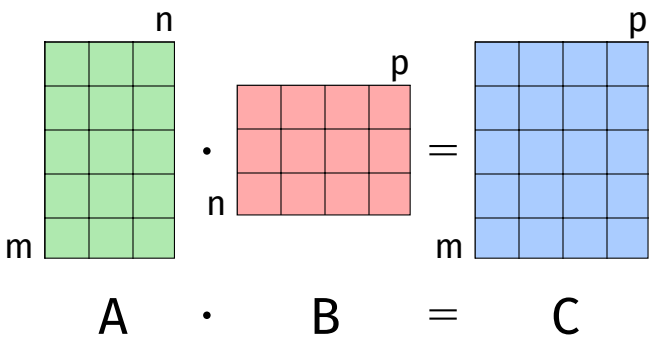
Given matrices $\boldsymbol{A} \in \mathbb{R}^{m \times n}$ and $\boldsymbol{B} \in \mathbb{R}^{n \times p}$, their product $\boldsymbol{C} = \boldsymbol{A}\boldsymbol{B}$ is:

$$C_{i,j} = \sum_{k=1}^{n} A_{i,k} B_{k,j}$$

Example: matrix-vector multiplication:

$$\begin{pmatrix} a_{1,1} & a_{1,2} \\ a_{2,1} & a_{2,2} \end{pmatrix} \cdot \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = \begin{pmatrix} a_{1,1} b_1 + a_{1,2} b_2 \\ a_{2,1} b_1 + a_{2,2} b_2 \end{pmatrix}$$

$$\underbrace{(2 \times 2)}_{\boldsymbol{A}} \cdot \underbrace{(2 \times 1)}_{\boldsymbol{b}} = \underbrace{(2 \times 1)}_{\boldsymbol{c}}$$



source: Wikipedia

Single-layer feed-forward neural network

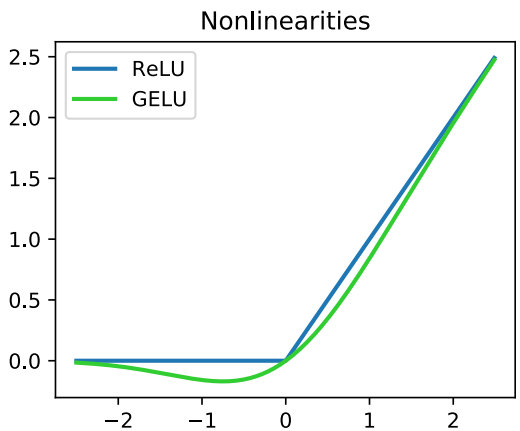
A single-layer **feed-forward neural network** (FFNN) applies a linear transformation (using the matrix $\boldsymbol{W}$ and bias $\boldsymbol{b}$) followed by a non-linear function σ (ReLU, GELU, ...):

$$\boldsymbol{y}_1 = \sigma(\boldsymbol{W}_1 \boldsymbol{x} + \boldsymbol{b}_1)$$

Example: FFNN with ReLU applied on a vector $\boldsymbol{x}$:

$$\underbrace{\begin{pmatrix} -0.38 & 1.35 \\ 0.70 & 0.30 \\ -1.03 & -1.03 \end{pmatrix}}_{\boldsymbol{W}_1 \in \mathbb{R}^{3\times 2}} \cdot \underbrace{\begin{pmatrix} -0.88 \\ 0.73 \end{pmatrix}}_{\boldsymbol{x} \in \mathbb{R}^2} + \underbrace{\begin{pmatrix} 0.10 \\ 0.21 \\ -0.48 \end{pmatrix}}_{\boldsymbol{b}_1 \in \mathbb{R}^3} = \underbrace{\begin{pmatrix} 1.42 \\ -0.19 \\ -0.33 \end{pmatrix}}_{\boldsymbol{z} \in \mathbb{R}^3}$$

$$\mathrm{ReLU}\left(\begin{pmatrix} 1.42 \\ -0.19 \\ -0.33 \end{pmatrix}\right) = \underbrace{\begin{pmatrix} 1.42 \\ 0.00 \\ 0.00 \end{pmatrix}}_{\boldsymbol{y}_1 \in \mathbb{R}^3}$$



source: Wikipedia

Softmax

Given a vector $x \in \mathbb{R}^d$, **softmax** ensures that $\forall x_i \geq 0$ and $\sum(x) = 1$:

$$\mathrm{softmax}(x_i) = \frac{\exp(x_i)}{\sum_{j=1}^{K} \exp(x_j)}$$

How to think about softmax

A convenient way how to transform any vector to a valid probability distribution.

Example:

$$z = [2.0, 1.0, 0.1]$$

$$\mathrm{softmax}(z) = [0.659, 0.243, 0.099]$$

Implementation in numpy:

```
def softmax(x):
    return np.exp(x) / np.sum(np.exp(x))
```

Why softmax?

Given a vector $x \in \mathbb{R}^d$, **softmax** ensures that $\forall x_i \geq 0$ and $\sum(x) = 1$:

```
def softmax(x):
    return np.exp(x) / np.sum(np.exp(x))
```

However, the same can be achieved with these two normalization functions:

```
def l1_norm(x):
    return np.abs(x) / np.sum(np.abs(x))
```

```
def min_sum_norm(x):
    x_norm = (x - np.min(x))
    return x_norm / np.sum(x_norm)
```

Question

What are the specifics and (dis)advantages of each of these three functions?

Tokenization

From words to tokens

Tokenization = splitting text into smaller units (**tokens**) that the model will process.

We assumed that our input is a **sequence of words**. That sounds good in theory, but worse in practice:

Question

How would you split these strings into meaningful units?

- `Pneumonoultramicroscopicsilicovolcanoconiosis`
- 大型语言模型
- 🤗❤️🍾🥳

Why subword tokenization?

Idea

Keep the common words, split rare words into subwords.

Compare:

- Word-level: `["unhappiness"]` → out of vocabulary
- Character-level: `["u","n","h","a","p","p","i","n","e","s","s"]`
- Subword: `["un", "happiness"]` ✓

Question

How to decide where to split the subwords?

Byte pair encoding (BPE)

source: Sennrich et al. 2016

Originally a data compression algorithm, adapted for NLP tokenization.

BPE algorithm

1. Start with a vocabulary of all **individual characters** in the training corpus.
2. Count all **adjacent pairs** of tokens in the corpus.
3. **Merge** the most frequent pair into a new token.
4. **Repeat** steps 2–3 until the desired vocabulary size is reached.

The result is a **merge table** that defines how to tokenize any new text.

📝 **Exercise time!**

BPE – step by step

Corpus: "aab baac aab baac caaba aac" → initial vocabulary: {a, b, c, _}

Step	Most frequent pair	New token
1	(a, a) → 6 times	aa
2	(aa, b) → 3 times	aab
3	(aa, c) → 3 times	aac
4	(aab, _) → 2 times	aab_

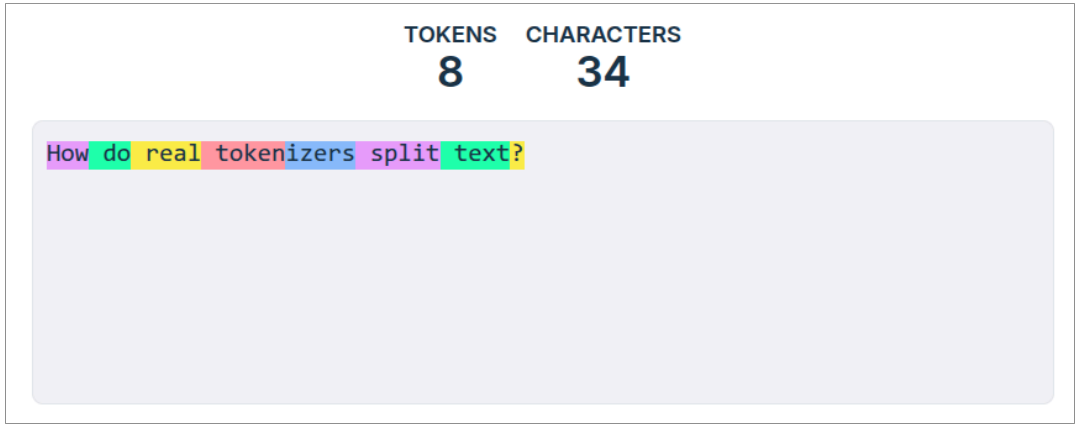
Final vocabulary: {_, a, b, c, aa, aab, aac, aab_}

Typical vocabulary sizes in practice: **50k–200k** tokens.

Tokenizer playground

How do real tokenizers split text?

👉 https://huggingface.co/spaces/Xenova/the-tokenizer-playground



Notice the difference between languages.

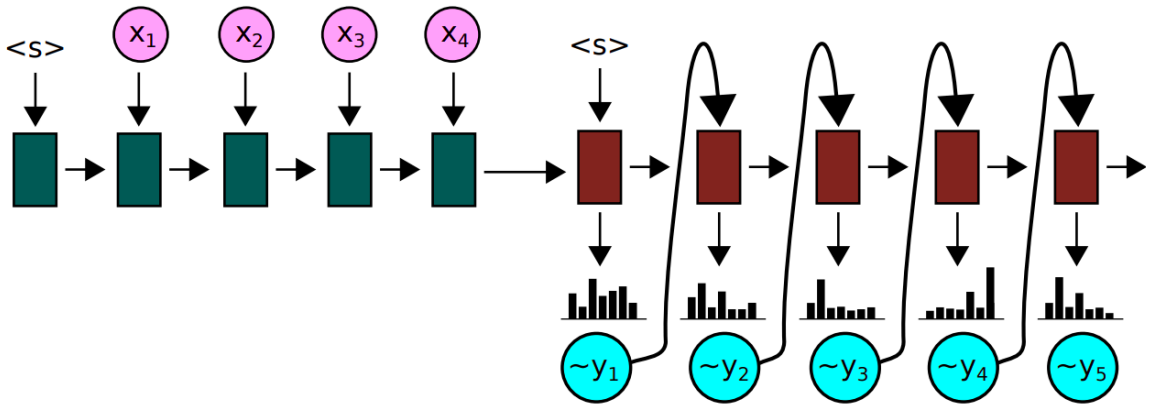
Attention mechanism

The seq2seq bottleneck

source: Sutskever et al. 2014

Remember: the encoder compresses the **entire input** into a single vector $\boldsymbol{h}_N$.

- Works OK for short sequences, but for longer inputs the **information gets lost**.
- The decoder has no way to "look back" at specific parts of the input.



source: NPFL116

Attention – the idea

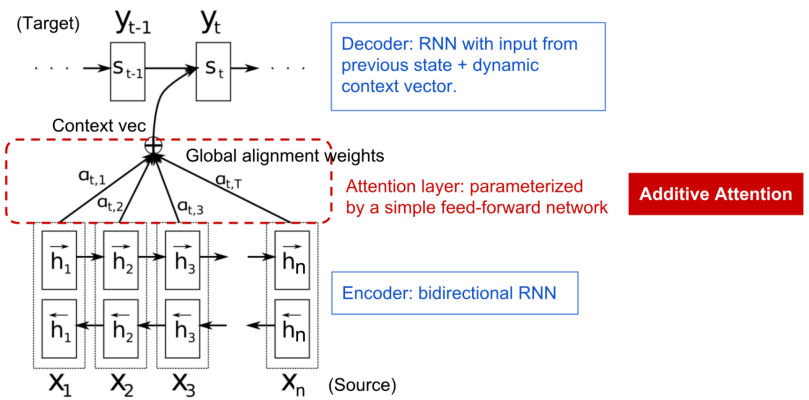
source: Bahdanau et al. 2015

Idea

Instead of compressing everything into one vector, let the decoder **attend to all encoder hidden states** at each step.

At each decoding step t:

1. Compute a **score** for each encoder state h_i.
2. Normalize scores with **softmax** → attention weights $\alpha_{t,i}$.



source: Lil'Log

3. Compute a **weighted sum** (context vector): $\boldsymbol{c}_t = \sum_i \alpha_{t,i} \boldsymbol{h}_i$
4. Use $\boldsymbol{c}_t$ together with the decoder state to predict the next token.

Bahdanau (additive) attention

source: Bahdanau et al. 2015

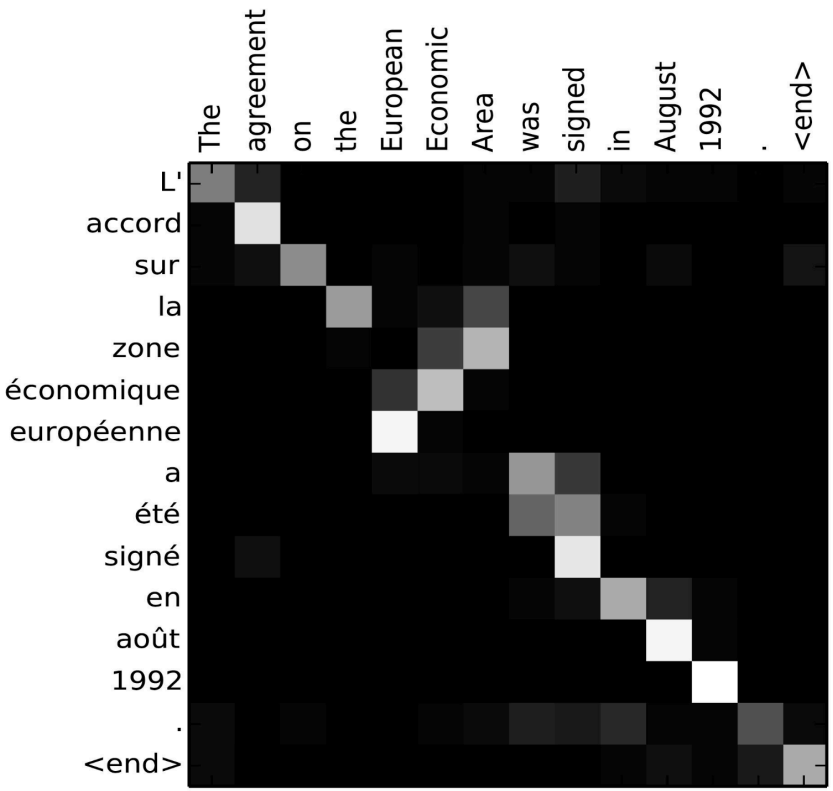
The score function uses a **learned feed-forward network**:

$$e_{t,i} = \boldsymbol{v}^\top \tanh(\boldsymbol{W}_1 \boldsymbol{s}_{t-1} + \boldsymbol{W}_2 \boldsymbol{h}_i)$$

$$\alpha_{t,i} = \mathrm{softmax}(e_{t,i}) = \frac{\exp(e_{t,i})}{\sum_j \exp(e_{t,j})}$$

$$\boldsymbol{c}_t = \sum_i \alpha_{t,i} \boldsymbol{h}_i$$

where $\boldsymbol{s}_{t-1}$ is the decoder state and $\boldsymbol{h}_i$ is the i-th encoder state.



source: Lil'Log

Luong (dot-product) attention

source: Luong et al. 2015

Simpler: use the **dot product** to compute the score:

$$e_{t,i} = \boldsymbol{s}_t^\top \boldsymbol{h}_i$$

or a **general** (bilinear) variant:

$$e_{t,i} = \boldsymbol{s}_t^\top \boldsymbol{W} \boldsymbol{h}_i$$

Key difference

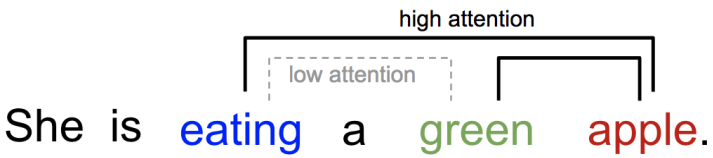
- **Bahdanau**: concat + learned layer → more expressive, slower.
- **Luong**: dot product → simpler, faster, and it turns out to work just as well.

Attention – visualized

The attention weights show **which input tokens the decoder focuses on** at each step.

For machine translation, the attention matrix often resembles a **soft alignment** between the source and target sentences.

This was a major breakthrough: finally, the model could **learn alignments on its own**, without any external alignment tool.



source: Lil'Log

From RNN attention to self-attention

So far, attention was used **between** the encoder and decoder (cross-attention).

Question

What if we use attention **within** a single sequence? Each word attending to every other word in the same sequence?

This is called **self-attention** – and it is the key building block of the Transformer.

Idea

And what if we get rid of the RNN entirely and use **only** attention?

The Transformer

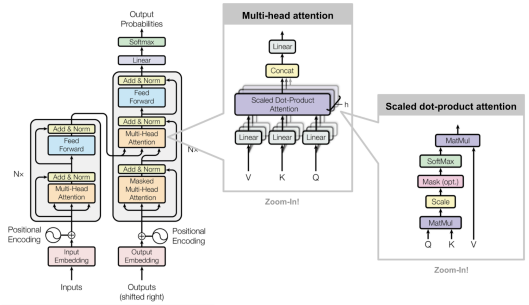
Attention is all you need

source: Vaswani et al. 2017

The paper that changed everything.

Key ideas:

- Replace RNNs entirely with **self-attention**.
- Process all tokens **in parallel** (no sequential bottleneck).
- Use **positional encoding** to inject position information.
- Stack multiple layers of attention + feed-forward networks.



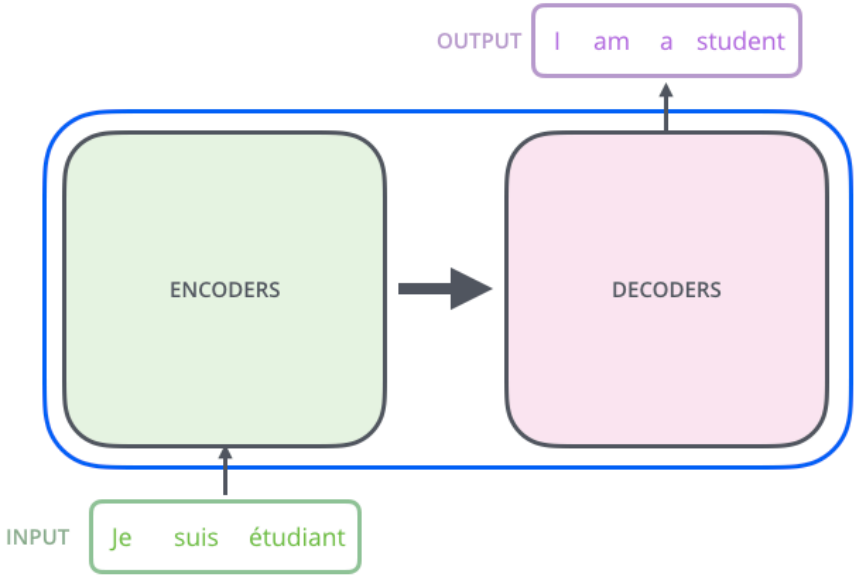
source: Lil'Log

Originally proposed for **machine translation**, but the architecture turned out to be universal.

Transformer – high-level view

source: The Illustrated Transformer

The Transformer follows the **encoder-decoder** pattern – same as seq2seq, but without any recurrence.



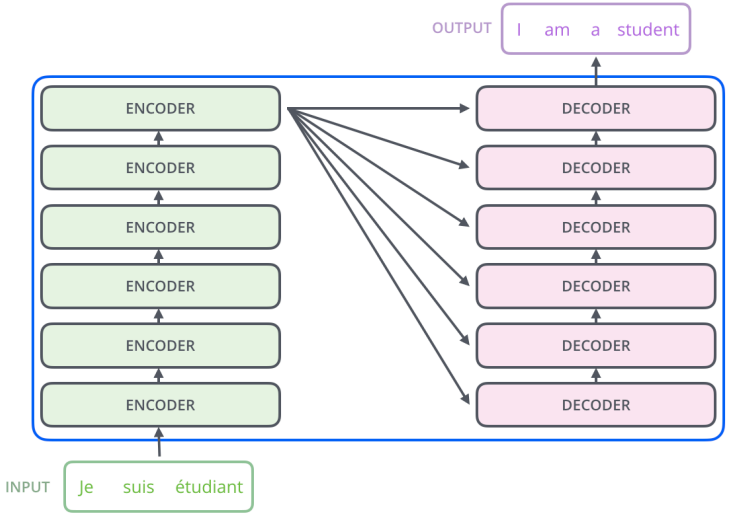
- **Encoder**: reads the input sequence, produces a rich representation.
- **Decoder**: generates the output sequence, one token at a time.

The encoder-decoder stack

source: The Illustrated Transformer

Both encoder and decoder are **stacks of identical layers** (blocks).

- The original Transformer uses $N = 6$ layers for both.
- Each layer refines the representation.
- The output of the last encoder layer is passed to **every** decoder layer.



Scaling up

Modern LLMs use many more layers: GPT-3 has 96, LLaMA-70B has 80.

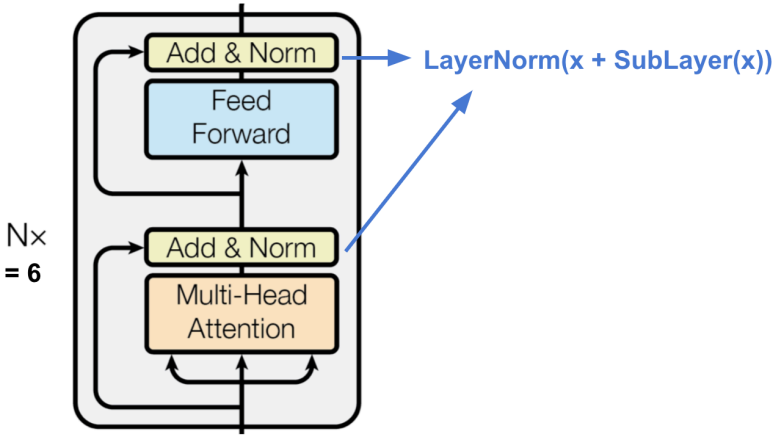
Inside an encoder block

source: The Illustrated Transformer

Each encoder block has two sub-layers:

1. **Self-attention layer**
   - Each token looks at all other tokens.
   - Captures dependencies regardless of distance.
2. **Feed-forward network (FFN)**
   - Applied to each token independently.
   - Same weights for every position.

Both sub-layers have a **residual connection** and **layer normalization**.



source: Lil'Log

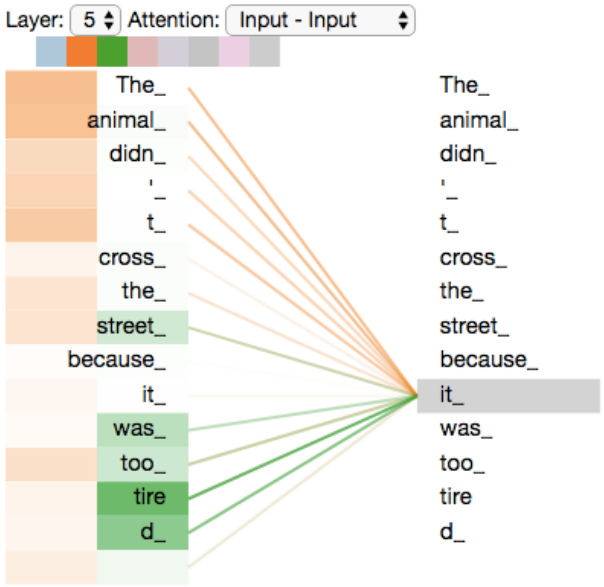
Self-attention – intuition

source: The Illustrated Transformer

The goal of self-attention: for each token, compute a **new representation** that incorporates information from **all tokens** in the sequence.

*"The animal didn't cross the street because **it** was too tired."*

When processing *"it"*, the model should attend strongly to *"animal"* (not *"street"*).

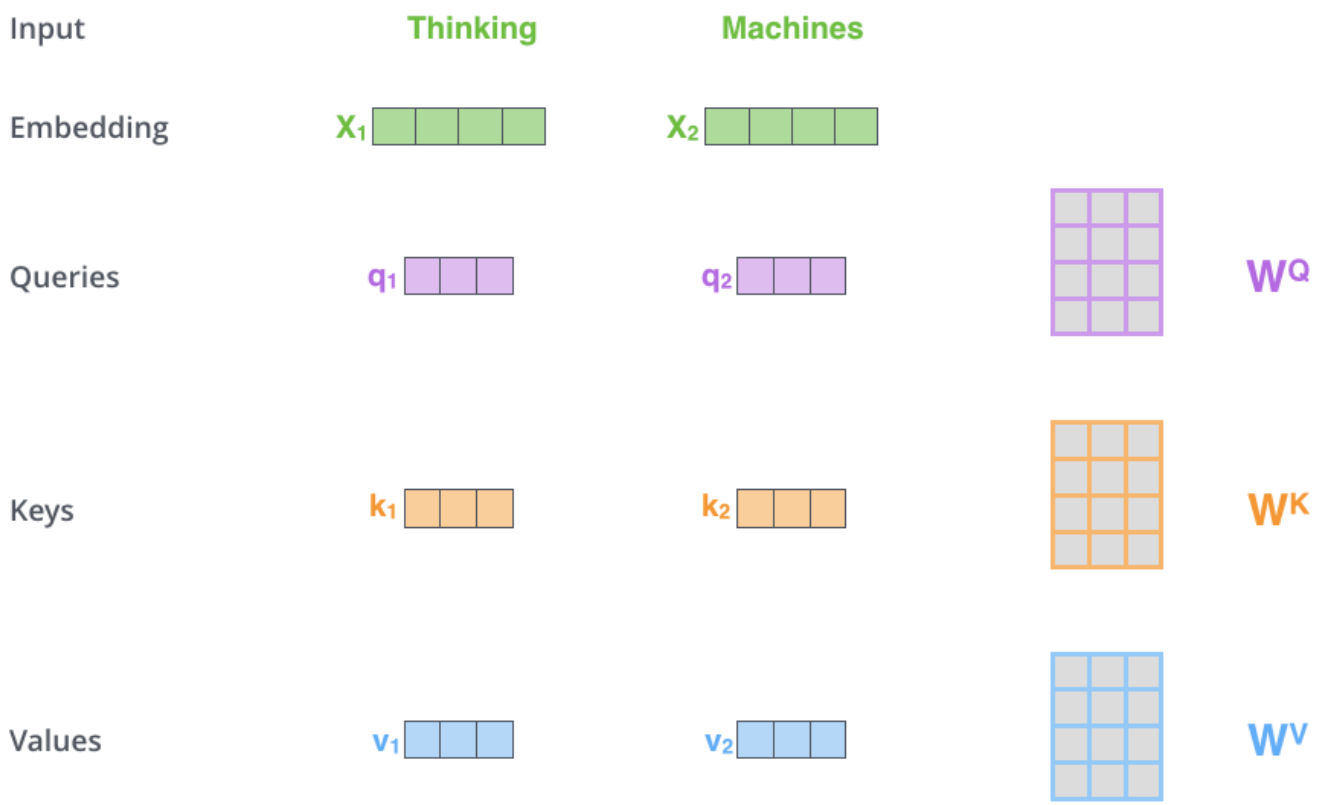


Self-attention – query, key, value

source: The Illustrated Transformer

For each token embedding $\boldsymbol{x}_i$, we compute three vectors:

$$\boldsymbol{q}_i = \boldsymbol{W}^Q \boldsymbol{x}_i, \quad \boldsymbol{k}_i = \boldsymbol{W}^K \boldsymbol{x}_i, \quad \boldsymbol{v}_i = \boldsymbol{W}^V \boldsymbol{x}_i$$


Input
Thinking
Machines
Embedding
X1
X2
Queries
q1
q2
WQ
Keys
k1
k2
WK
Values
v1
v2
WV

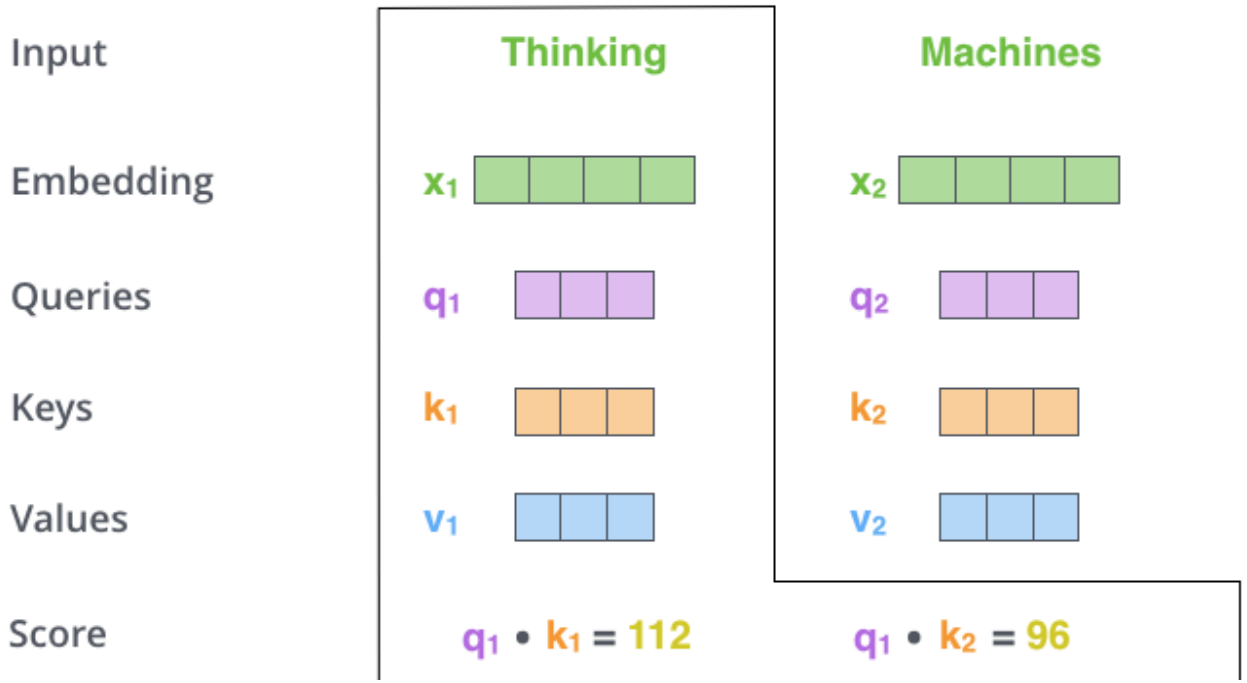
- **Query** (q): “what am I looking for?”
- **Key** (k): “what do I contain?”
- **Value** (v): “what information do I provide?”

Self-attention – computing scores

source: The Illustrated Transformer

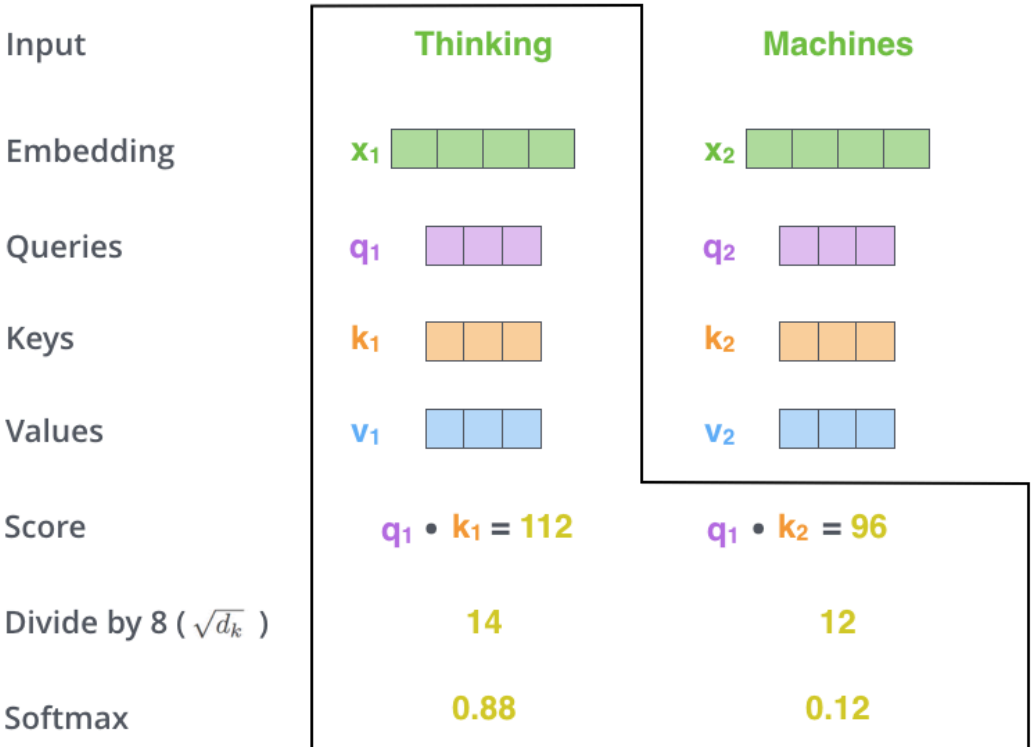
For a given query q_i:

1. Compute the **dot product** $q_i \cdot k_j$ for all keys → raw scores.
2. **Scale** by $\frac{1}{\sqrt{d_k}}$ (to keep gradients stable).
3. Apply **softmax** to get attention weights.
4. **Weighted sum** of value vectors.


Input
Thinking
Machines
Embedding
x1
x2
Queries
q1
q2
Keys
k1
k2
Values
v1
v2
Score
q1 • k1 = 112
q1 • k2 = 96

Self-attention – softmax and output

source: The Illustrated Transformer



The output for token i is a **weighted mixture** of all value vectors. Tokens with higher attention weights contribute more.

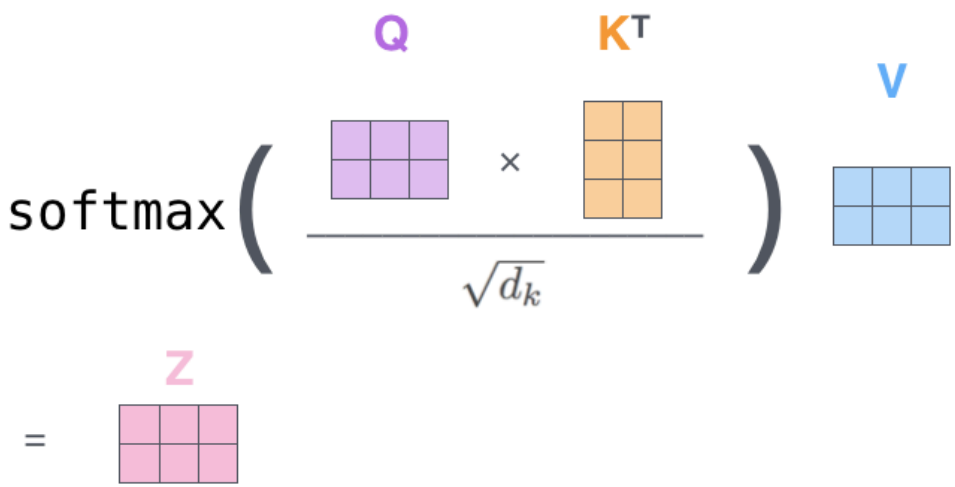
Scaled dot-product attention – matrix form

source: Vaswani et al. 2017

In practice, we compute everything in **one matrix operation**:

$$\mathrm{Attention}(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) = \mathrm{softmax}\left(\frac{\boldsymbol{Q}\boldsymbol{K}^\top}{\sqrt{d_k}}\right)\boldsymbol{V}$$

where $\boldsymbol{Q} \in \mathbb{R}^{n \times d_k}$, $\boldsymbol{K} \in \mathbb{R}^{n \times d_k}$, $\boldsymbol{V} \in \mathbb{R}^{n \times d_v}$.


Q
K^T
V
softmax
×
$\sqrt{d_k}$
Z
=

Multi-head attention – why?

source: Vaswani et al. 2017

A single attention head captures **one type of relationship**. But words relate to each other in many ways:

- syntactic (subject–verb),
- semantic (co-reference),
- positional (adjacent words),
- …

Idea

Run **multiple attention heads in parallel**, each with its own $\boldsymbol{W}^Q$, $\boldsymbol{W}^K$, $\boldsymbol{W}^V$ – and let each head learn a different pattern.

Multi-head attention – how it works

source: The Illustrated Transformer

With h heads and model dimension d:

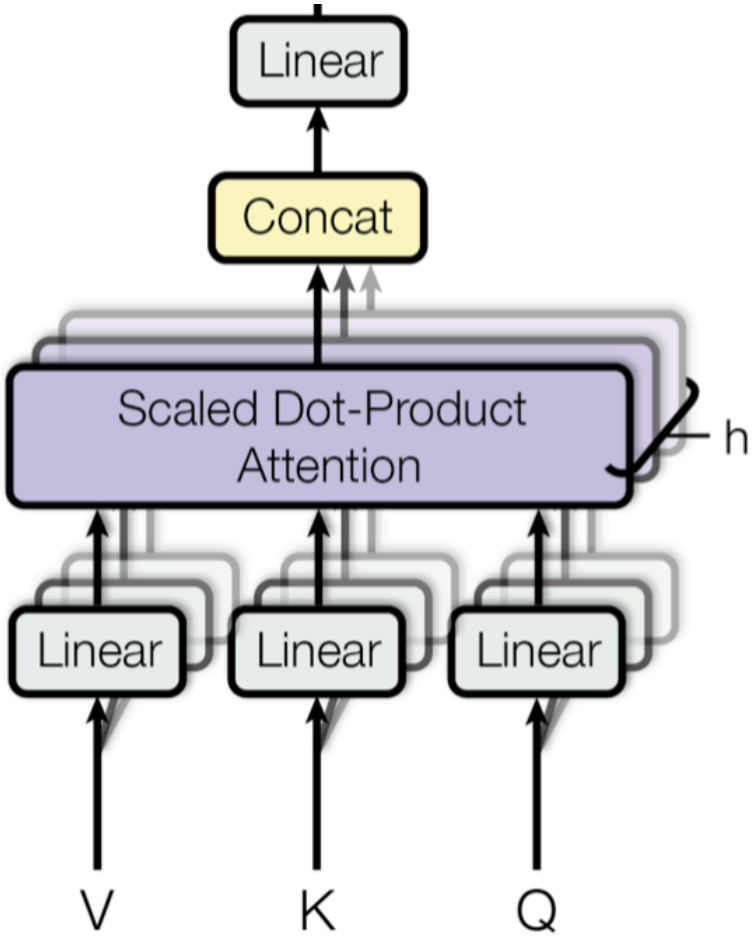
- Each head uses $d_k = d_v = \frac{d}{h}$.
- Each head computes its own Q, K, V and produces an output $\boldsymbol{Z}_i$.

$$\text{head}_i = \text{Attention}(\boldsymbol{Q}_i, \boldsymbol{K}_i, \boldsymbol{V}_i)$$

The heads are **concatenated** and projected:

$$\text{MultiHead} = [\boldsymbol{Z}_1; ...; \boldsymbol{Z}_h]\boldsymbol{W}^O$$

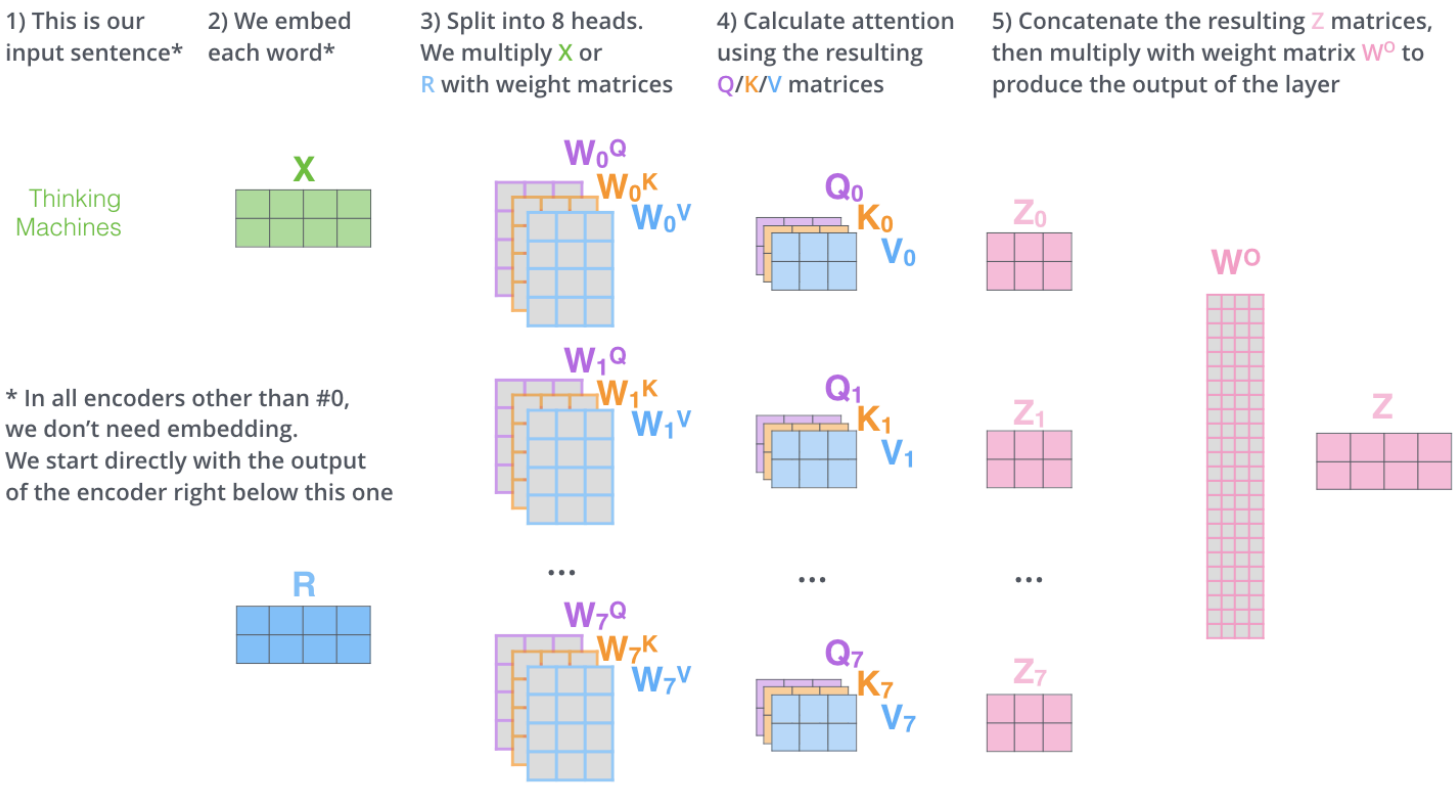
where $\boldsymbol{W}^O \in \mathbb{R}^{hd_v \times d}$.



source: Lil'Log

Multi-head attention – visualization

source: The Illustrated Transformer



Each head produces its own attention output → they are concatenated → multiplied by $\boldsymbol{W}^O$ to get the final output.

Residual connections & layer normalization

source: The Illustrated Transformer

Each sub-layer (attention or FFN) is wrapped in:

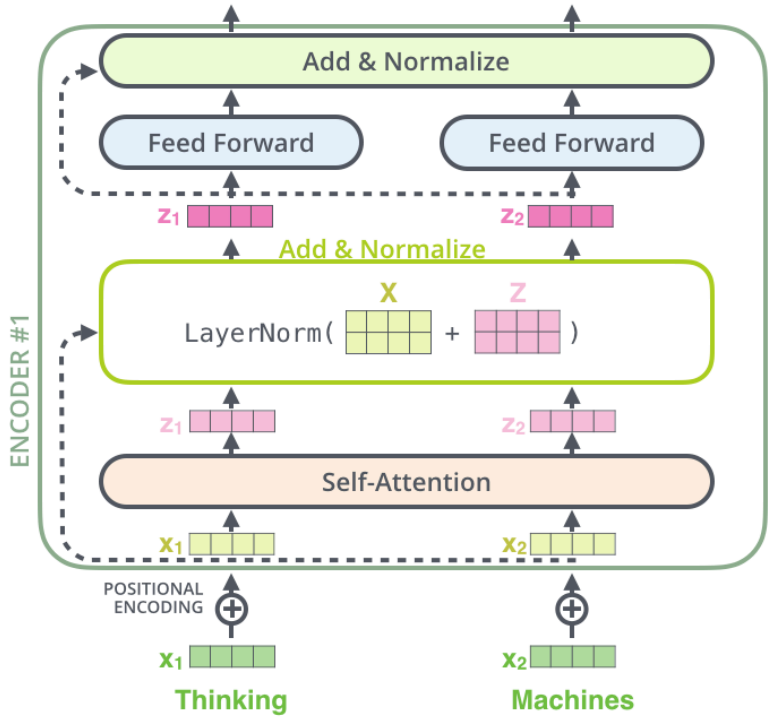
$$\mathrm{LayerNorm}(\boldsymbol{x} + \mathrm{SubLayer}(\boldsymbol{x}))$$

Residual connection ($x + \ldots$):

- Helps gradients flow through deep networks.
- The layer only needs to learn the **"delta"**.

Layer normalization:

- Normalizes across features (not across batch).
- Stabilizes training.



Position-wise feed-forward network

Applied to **each token independently** (same weights, different inputs):

$$\mathrm{FFN}(\boldsymbol{x}) = \boldsymbol{W}_2 \cdot \mathrm{ReLU}(\boldsymbol{W}_1 \boldsymbol{x} + \boldsymbol{b}_1) + \boldsymbol{b}_2$$

where $\boldsymbol{W}_1 \in \mathbb{R}^{d \times d_{\mathrm{ff}}}$, $\boldsymbol{W}_2 \in \mathbb{R}^{d_{\mathrm{ff}} \times d}$.

Dimensions

In the original Transformer:

- $d = 512$ (model dimension)
- $d_{\mathrm{ff}} = 2048$ (inner dimension, 4× larger)

What does the FFN do?

No interaction between positions – that is the job of self-attention. The FFN processes each token **on its own**, acting as a **per-token memory** storing factual knowledge.

Positional encoding – why do we need it?

Self-attention is **permutation-invariant**: if we shuffle the input tokens, the attention scores would be the same (up to the same permutation).

Question

Why is that a problem? *"The cat sat on the mat"* and *"mat the on sat cat the"* should be different!

Idea

We need to **inject position information** into the input embeddings so that the model can distinguish token order.

Positional encoding – sinusoidal

source: Vaswani et al. 2017

The original Transformer uses **fixed sinusoidal functions**, added to the token embeddings:

$$\boldsymbol{x}_i = \mathrm{Embed}(w_i) + \mathrm{PE}(i)$$

$$\mathrm{PE}(p, 2i) = \sin\left(\frac{p}{10000^{2i/d}}\right), \quad \mathrm{PE}(p, 2i+1) = \cos\left(\frac{p}{10000^{2i/d}}\right)$$

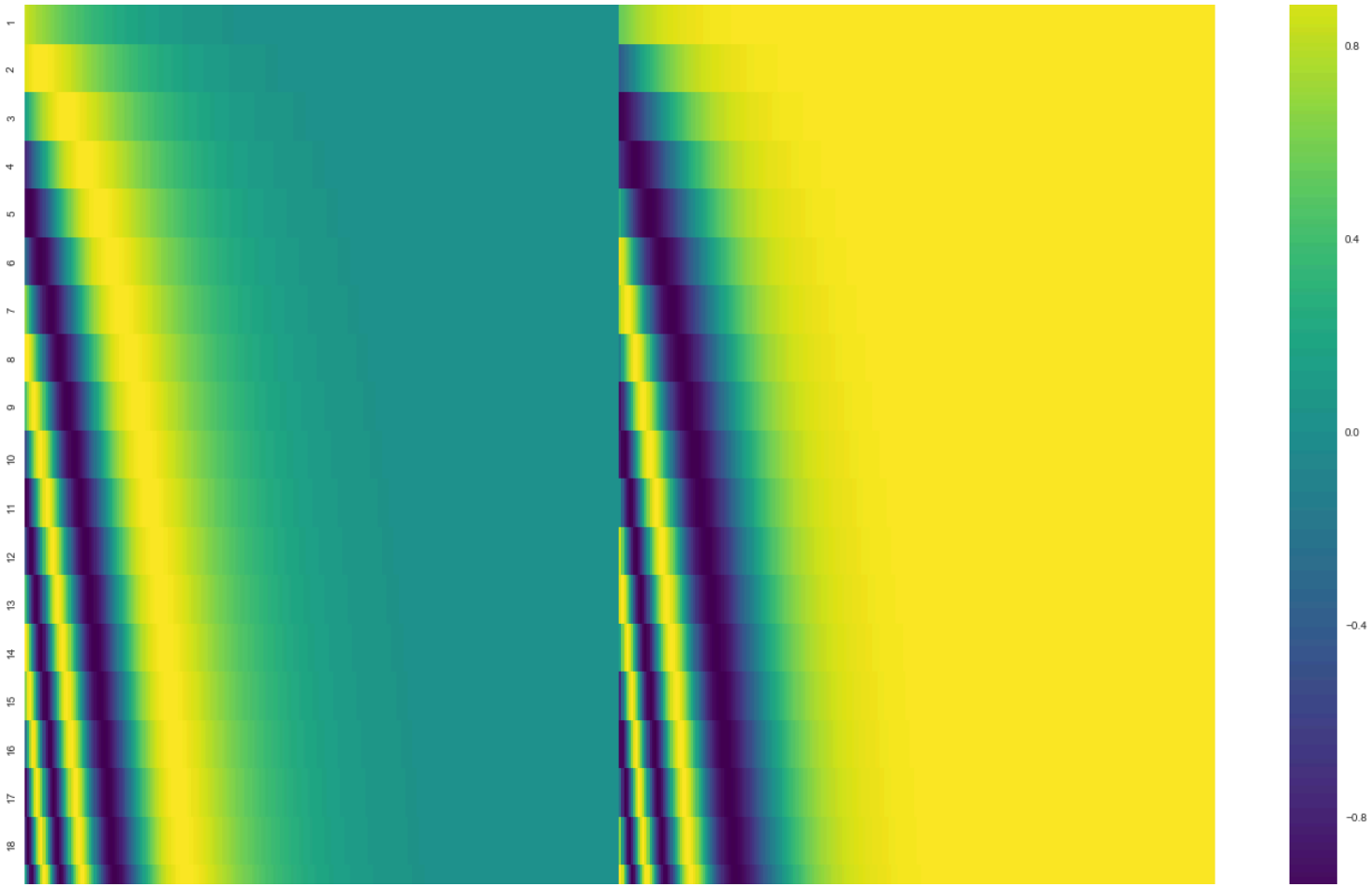
where p is the position and i is the dimension index.

- Each position gets a **unique encoding**.
- Nearby positions have **similar encodings**.
- The model can learn to attend to **relative positions**.

Modern models often use **learned** positional embeddings or **rotary positional encoding** (RoPE) instead.

Positional encoding – visualization

source: The Illustrated Transformer



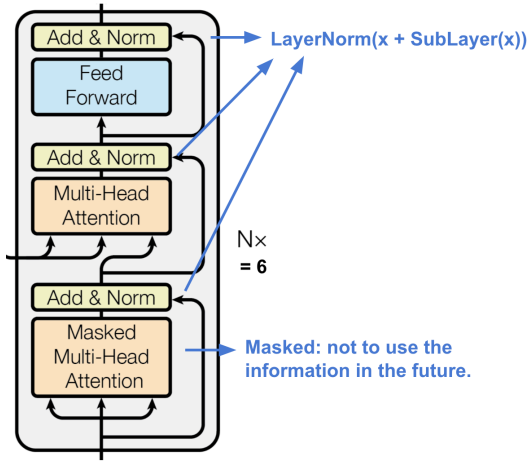
Each row is a positional encoding vector. The pattern changes across dimensions: low-frequency sinusoids on the left, high-frequency on the right.

The decoder

source: The Illustrated Transformer

The decoder has **three sub-layers** per block:

1. **Masked self-attention**
   - Each token can only attend to previous tokens (no peeking into the future).
2. **Encoder-decoder (cross) attention**
   - Queries come from the decoder, keys and values come from the encoder output.
3. **FFN**
   - Same as in the encoder.



source: Lil'Log

Masked self-attention

source: Vaswani et al. 2017

During training, we know the full target sequence. But we must prevent the decoder from **looking ahead**.

We apply a **mask** to the attention scores before softmax:

$$\mathrm{MaskedAttention}(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) = \mathrm{softmax}\left(\frac{\boldsymbol{Q}\boldsymbol{K}^\top}{\sqrt{d_k}} + \boldsymbol{M}\right)\boldsymbol{V}$$

where $\boldsymbol{M}$ is a matrix with 0 for allowed positions and $-\infty$ for future positions.

Effect

$\exp(-\infty) = 0$ → after softmax, future tokens get zero attention weight. Token t can only attend to tokens $1, \ldots, t$.

Encoder-decoder attention

In the cross-attention sub-layer:

- **Queries** come from the decoder (previous sub-layer).
- **Keys** and **values** come from the **encoder output**.

$$\mathrm{CrossAttention}(\boldsymbol{Q}_{\mathrm{dec}}, \boldsymbol{K}_{\mathrm{enc}}, \boldsymbol{V}_{\mathrm{enc}})$$

This allows every decoder position to attend over **all positions in the input** – just like the attention mechanism in the RNN seq2seq, but more powerful.

> **This is the bridge**

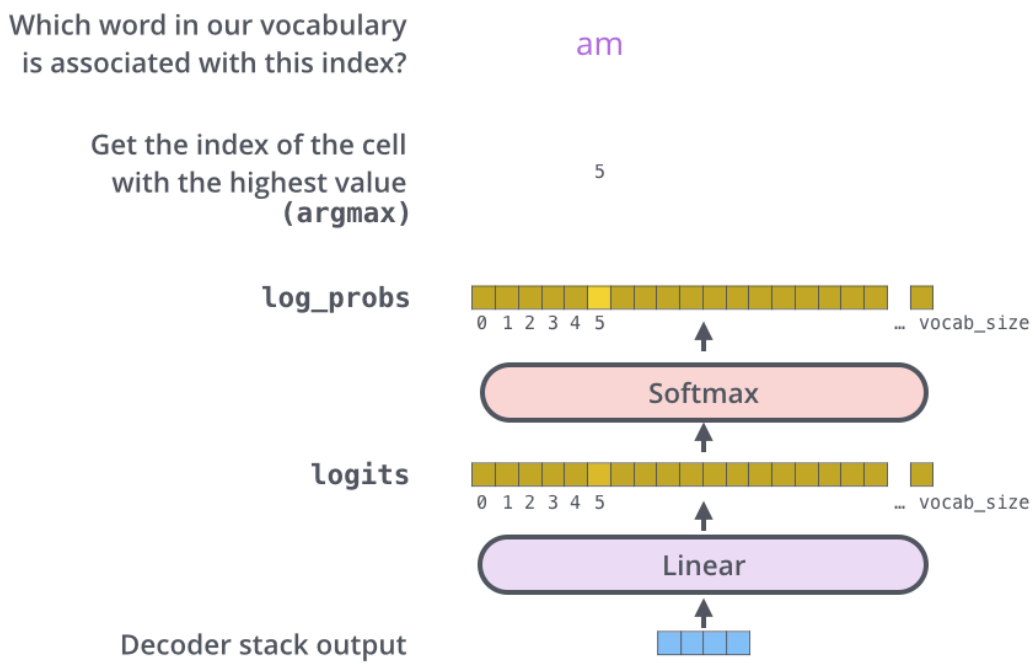
> Cross-attention is what connects the encoder and decoder. Without it, the decoder would have no access to the source sequence.

Output layer

source: The Illustrated Transformer

The final decoder output is projected to vocabulary size and passed through **softmax**:

$$P(w_t \mid w_1, ..., w_{t-1}, \boldsymbol{x}) = \mathrm{softmax}(\boldsymbol{W}_{\mathrm{out}} \boldsymbol{h}_t + \boldsymbol{b})$$



Same idea as neural language modeling from Lecture 1 – but now the hidden state h_t is much richer thanks to attention.

Training the Transformer

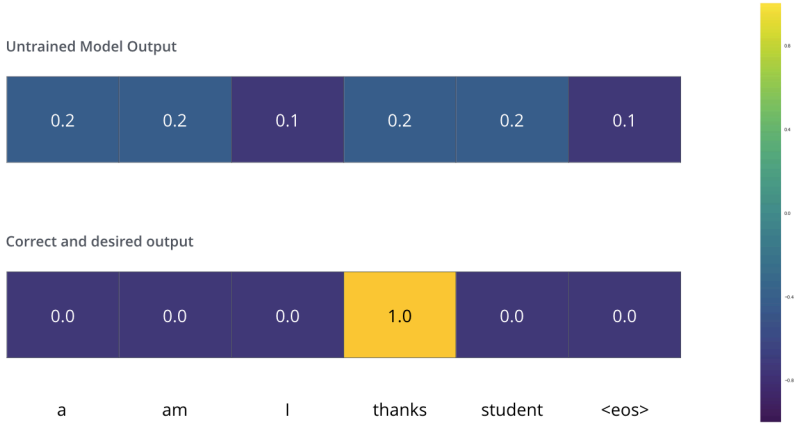
source: The Illustrated Transformer

Objective: maximize the probability of the correct target tokens.

→ **Cross-entropy loss** between predicted distribution and one-hot target:

$$\mathcal{L} = -\sum_{t=1}^{T} \log P(w_t^* \mid w_1^*, \ldots, w_{t-1}^*, \boldsymbol{x})$$

Thanks to masking, we can compute the loss for **all positions in parallel** during training (unlike RNNs!).



→ **Teacher forcing**: use the ground-truth tokens as decoder input.

Why is the Transformer better than RNNs?

RNNs

- Sequential processing → **slow**.
- Information must pass through **all intermediate states**.
- Gradient vanishing over long distances.
- Hidden state is a **bottleneck**.

Transformer

- Parallel processing → **fast**.
- Any token can directly attend to **any other token** (path length = 1).
- Stable gradients with residual connections.
- No bottleneck: information distributed across **all positions**.

Computational cost

Self-attention is $O(n^2 \cdot d)$ per layer. For very long sequences, this can be expensive – but there are efficient attention variants (→ later lectures).

Transformer variants

The original Transformer is encoder-decoder. But there are **three main variants**:

Encoder-only

BERT, RoBERTa

- Bidirectional context.
- Good for **classification**, **NER**, **retrieval**.

Decoder-only

GPT, LLaMA, Gemma

- Autoregressive (left-to-right).
- Good for **text generation**, **language modeling**.

Encoder-decoder

T5, BART, mBART

- Full seq2seq.
- Good for **translation**, **summarization**.
- Less common now in large-scale LLMs.

- Cannot generate text.

- The dominant LLM architecture today.

The decoder-only Transformer

Most modern LLMs (GPT-4, Claude, LLaMA, ...) are **decoder-only**:

- No encoder, no cross-attention.
- The input and output are **one single sequence**.
- Uses **masked (causal) self-attention** throughout.

For each position t, the model predicts the next token:

$$P(w_{t+1} \mid w_1, ..., w_t)$$

At inference, we generate tokens **autoregressively** – feeding each generated token back as input.

Why decoder-only?

- Simpler architecture.
- Scales well.
- Unifies many tasks under "generate the answer".

Summary

Summary

What we covered today

- **Refresher**: matrix multiplication, FFNNs, softmax.
- **Tokenization**: subword tokenization with BPE.
- **Attention**: solving the seq2seq bottleneck (Bahdanau, Luong).
- **The Transformer**:
  - Self-attention with Q, K, V → scaled dot-product attention.
  - Multi-head attention → multiple parallel attention heads.
  - Positional encoding → injecting position information.
  - Residual connections + layer normalization.

- Encoder-decoder architecture + decoder-only variants.

Next time: pre-training objectives, training data, finetuning, RLHF.

Links

- The Illustrated Transformer – Jay Alammar
- Attention? Attention! – Lil'Log
- The Annotated Transformer – Harvard NLP
- Attention is all you need – Vaswani et al. 2017
- Neural machine translation by jointly learning to align and translate – Bahdanau et al. 2015
- Neural machine translation of rare words with subword units – Sennrich et al. 2016
- Tokenizer playground – Hugging Face